

Dr. BHAWAKSHI PUNIA

AI & Computational Scientist | PhD



PROFESSIONAL EXPERIENCE

[Aganitha.ai](#)

March 2025 - Present

Jr. AI & Computational Scientist 2 | March 2026 - Present

- Led client-focused **end-to-end ML development for CMC prediction of surfactant mixtures** in homecare products, spanning data curation/cleaning, model benchmarking, featurisation evaluation, and delivery of a user-friendly prediction interface.

Jr. AI & Computational Scientist 1 | September 2025 - February 2026

- Built end-to-end **AI/ML pipelines (PyTorch/MXNet)** for client projects, making predictions via **GNNs** and fine-tuned **foundational models**.
- Delivered classifiers for nitrosation impurity in drug APIs & ionic liquid viscosity for subcutaneous injection formulations and CMC regression model for surfactant mixtures.

AI & Computational Scientist Intern | March 2025 - August 2025

- Collaborated on cross-disciplinary enzyme engineering project using **ML/DL** and **Protein Language Models** for predictive pipelines.
- Implemented **ClearML** for experiment tracking, iterative development, and performance evaluation.

SELECTED PROJECTS

Sweet Truth Detector: Identifying Hidden Sweeteners in Packaged Foods </>

Built an ML pipeline for sweetener identification from text/images (97% text accuracy) using Tesseract OCR, OpenCV, and spaCy; delivered results as annotated images.

Predicting pEC50 for Dopamine Receptors </>

Curated a ChEMBL dataset with RDKit features & developed prediction model achieving MAE = 0.60.

EDUCATION

Ph.D. Theoretical Chemistry | Indian Institute of Science Education and Research (IISER) Pune, India

M. Sc. Chemistry | Miranda House, University of Delhi, India

B. Sc. (Hons.) Chemistry | Sri Venkateswara College, University of Delhi, India

ADDITIONAL INFORMATION

Achievements

- Best Poster Award** at International Conference on Biological Physics (ICBP), **Seoul, Korea** - Aug 2023
- Best Poster Award** at Designing Catalysts on Computers (IACS Kolkata, India) - Dec 2022
- Best Oral Presentation** at CSIR NCL-RF: Annual Students' Conference - Nov 2022

Technical skills

Python for data science • MLOps (Docker) • ClearML • FORTRAN • LaTeX